# Masters of Design in Intelligent User Experience Design – Mdes (IUxD)

## Semester -2

**MI611:**

**Design Thinking: Concepts & concerns**

**Credit structure: 2-0-2-3**

**Content:** This course explores Design Thinking as a structured, human-centered approach to problem-solving, fostering creativity and strategic innovation. Students will engage with key methodologies, including stakeholder mapping, creative matrix, schematic diagramming, and problem tree analysis, to develop solutions that address real-world challenges. The course also examines the creative economy and the role of design in business strategy. By applying design thinking principles, students will produce system design and decision-making (DM) plans and craft compelling product/service pitches using business canvas models and go-to-market strategies. Through hands-on exercises and case studies, students will develop critical thinking, collaboration, and problem-solving skills essential for applying design thinking in various professional contexts. This course equips students with practical tools to drive innovation and create meaningful user-centered solutions.

**Course Structure: Module submission: 60%**
**End Term Jury: 40%**

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**MI612:**

**Design Research Methods**

**Credit structure: 2-0-2-3**

**Content:** This course provides a comprehensive understanding of design research, emphasizing both quantitative and qualitative methodologies. Students will explore key UX research techniques, including diary studies, desk research, ethnographic research, surveys, field studies, card sorting, contextual inquiry, tree testing, and focus group interviews. The course equips students with essential research tools, enabling them to systematically engage with users, experts, and environments to gather valuable insights. By learning structured research approaches, students will develop the ability to analyze data effectively, identify user needs, and inform design decisions. Practical application of these methods will help them produce well-structured research reports that support user-centered and evidence-based design solutions. By the end, students will be proficient in conducting rigorous research, ensuring that design outcomes are both meaningful and impactful.

**Course Structure: Module submission: 60%**
**End Term Jury: 40%**

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**MI613**

**AI & Interaction Intelligence**

**Credit structure: 2-0-4-4**

**Content:** This course explores the fundamentals of Artificial Intelligence (AI), focusing on its concepts, capabilities, and concerns. Students will gain an understanding of how AI models function, the types of data required, and their impact on design and interaction. A key focus is on Generative AI (GenAI) and its applications in creative problem-solving and design innovation. The course also introduces interaction intelligence, examining how AI enables conversations between objects and humans. Through conceptual models and hands-on exploration, students will learn to integrate AI into design workflows, enhancing user experiences and interaction design. By the end, they will have a strong foundation in AI-driven design thinking, enabling them to conceptualize and develop intelligent, user-centered solutions.

**Course Structure: Module submission: 60%**
**End Term Jury: 40%**

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**MI614:**

**Intelligent Design Decisions**

**Credit structure: 2-0-4-4**

**Content:** This course explores the principles of information design and data visualization, focusing on transforming complex data into meaningful and user-centric representations. Students will learn to analyse, categorize, and visualize data in context to stakeholders, ensuring clarity and usability. The course covers data collection methods, representation techniques using tools like Tableau, and cognitive walkthroughs to assess user interaction with information. Additionally, students will explore infographics, signages, and wayfinding systems to enhance communication. Through hands-on exercises, they will develop skills in structuring data for effective storytelling and decision-making. By the end, students will understand the concept of data manifestation and be able to create visually compelling, insightful, and functional information designs that enhance user experiences.

**Course Structure:** **Module submission: 60%**
**End Term Jury: 40%**

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**MI615:**

**Human Factors (Physical and Cognitive Ergonomics) in Interaction Design**

**Credit structure: 2-0-2-3**

**Content:** This course explores the critical role of human factors in designing user-friendly digital products and interfaces. Students will learn about physical and cognitive ergonomics, focusing on how users interact with technology in terms of comfort, efficiency, utility, and safety. Key topics include cognitive load management, anthropometry in digital products, and the impact of human factors on display design, communication, and behavioural design. The course also examines how human-centered design reduces errors and enhances user experiences. Through a redesign process informed by ergonomics, students will apply their knowledge to create intuitive, accessible, and efficient digital solutions. By the end, they will develop a deep understanding of how human factors influence interaction design, enabling them to craft more effective and user-centric interfaces.

**Course Structure:** **Module submission: 60%**
**End Term Jury: 40%**

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**MI616:**

**User Interface Design**

**Credit structure: 2-0-4-4**

**Content:** This course provides a comprehensive understanding of User Interface (UI) design, focusing on creating intuitive and intelligent digital interactions. Students will explore the AEIOU framework for user analysis, core UI design principles, and platform-specific considerations for mobile, web, and kiosks. The course covers essential concepts such as Information Architecture (IA), user journey mapping, and wireframing across low, medium, and high fidelity. Additionally, students will engage in A/B testing to optimize user experiences and refine digital interfaces. Emphasis is placed on crafting seamless digital experiences through structured grids, layouts, and style guides. By the end, students will be able to design, prototype, and evaluate user-friendly interfaces, producing wireframes, IA structures, and interactive digital prototypes that align with user needs and industry standards.

**Course Structure:** **Module submission: 60%**
**End Term Jury: 40%**

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